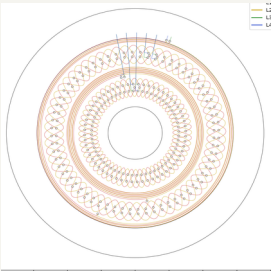
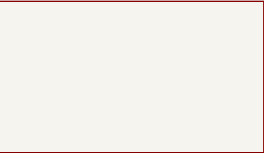


INDUCTIVE ENCODER – STATOR (COIL) BOARD (rev A)

4-layer 76 mm stator with the dual-track Nonius coils (16 / 15 periods, 58 mm coil area, 16 mm shaft hole). The target disc rotates 0.5 mm above L1. The signal board (signal/EncoderSignal) stacks behind it on M3 standoffs and plugs into J2/J3. No GND copper on this board: a closed ring around the excitation coils would act as a shorted turn.

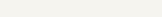


Coils

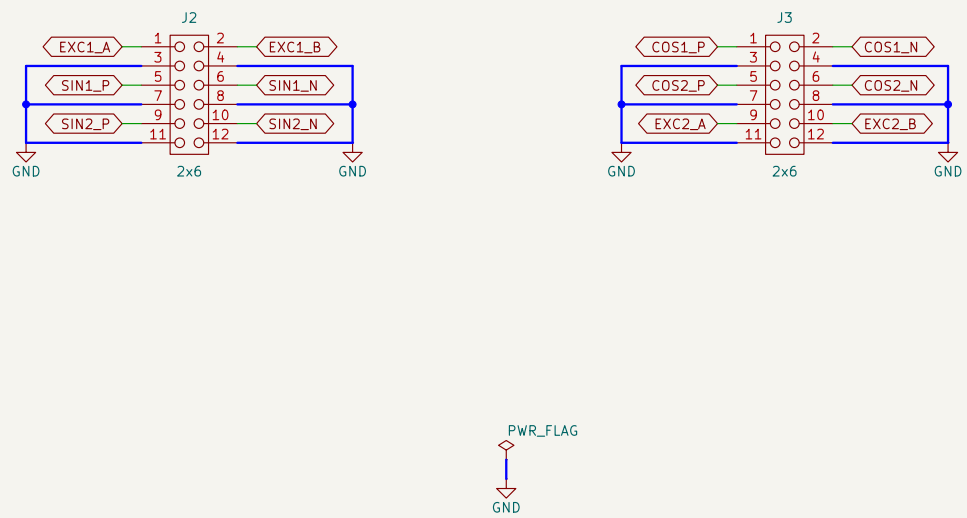


File: coils.kicad_sch

REVISION HISTORY
A 2026-10-09 first issue

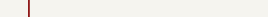
Title: Inductive Encoder – Stator (coils)			
Project: Inductive Encoder			
Number: TBD		Rev: A	
Sheet: /		File: InductiveEncoder.kicad_sch	
Size: A4	Date: 2026-10-09	Designer : Ekrem Turan Firat	

COILS (PCB copper generated by tools/coilgen.py)



Each coil is drawn as a net tie: the coil copper and both of its leads carry the _P/_A net; the tie at the end of the N lead joins the _N/_B net.
LAYOUT: net ties at the coil lead exits; coil terminals are hand-routed (tools/stator_layout.py) because both lead ends belong to the same net.

J2/J3: 2x6 2.54 mm male headers on the BOTTOM side at the coil exits.
Mate with the signal-board sockets J2/J3 (same pinout). GND comes from the signal board; it only shields the pairs inside the connector.

Title: Inductive Encoder – Stator (coils)				
Project: Inductive Encoder				
Number: TBD	Rev: A			
Sheet: /Coils/	File: coils.kicad_sch			
Size: A4	Date: 2026–10–09		Designer : Ekrem Turan Firat	